

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Valve Spring Compressor Kit, Part Number 4919358
- Boot Pliers, Part Number 3163293
- Valve Seal Installer, Part Number 3163294.

Additional Service Items

- No additional service items required.

General Information

Three types of valve guide seals may be installed on ISX15 and QSX15 engines.

Seals 1 and 2 fit directly on the top of the valve guide and were used on engines assembled prior to 11 Feb 2015.

Seal 3 fits over the valve guide with an integrated valve spring guide and have been assembled in engines since 11 Feb 2015.

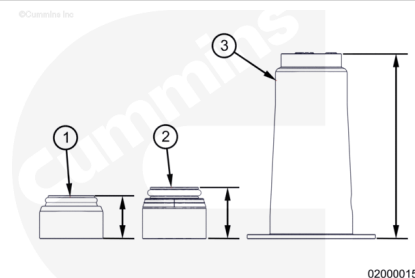
If seal 1 malfunctions, then **only** the seals that have malfunctioned **must** be replaced.

If seal 2 malfunctions, then all seals **must** be replaced.

If seal 3 malfunctions, then **only** the seals that have malfunctioned **must** be replaced.

Note : Seal 1 is 12.85 mm [0.505 in] from the bottom of the metal casing to the top of the rubber seal. Seal number 2 is 14.25 mm [0.561 in] from the bottom of the metal casing to the top of the rubber seal.

Note : Seal 3 is identifiable by an integrated valve spring guide. The seals are **not** reusable and **must** be discarded if removed for any reason.



Preparatory Steps

On-Engine Repair

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

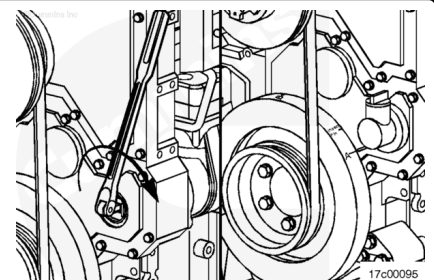
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the coolant to below the cylinder head level. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>)
- Disconnect the engine brake solenoid wiring harness. Refer to Procedure 020-015 in Section 20. (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>)
- Remove the injector for the cylinder to be repaired. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html>)
- Remove the rocker lever shaft assemblies. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>).

Bar the engine over so the cylinder being worked on is at top dead center (TDC). Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html>)

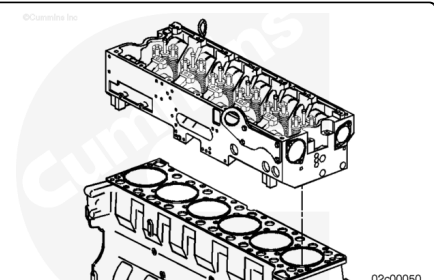
Repeat this step as needed for each cylinder.



Off-Engine Repair

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

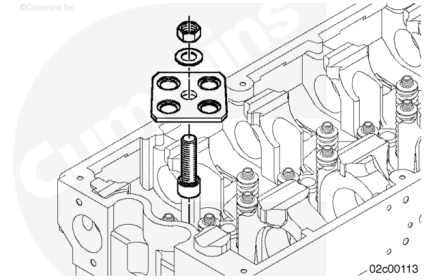


- Remove the cylinder head and set it on a bench or table.
Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)

Remove

⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if improperly released. To reduce the possibility of eye injury, wear safety glasses with side shields. Before releasing the valve spring tension, check that the collets are properly seated.



⚠ CAUTION ⚠

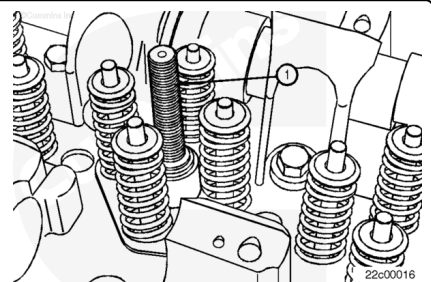
**Do not use an impact tool for any part of this procedure.
Doing so can damage the tool and/or the engine.**

Note : This step can be performed either on- or off-engine.

Use four valve spring compressor kit, Part Number 4919358, to compress the valve springs.

Remove the spring retainer collets.

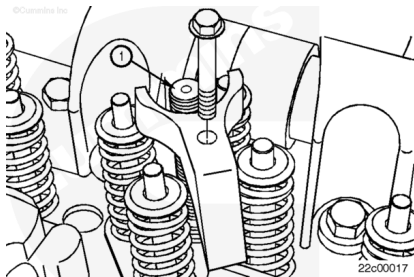
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

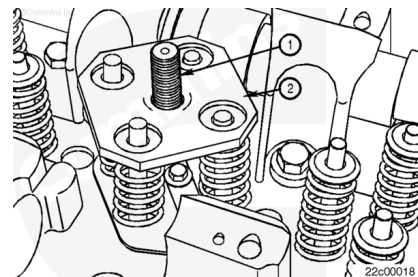
Tighten the capscrew.

Torque Value: 54 n•m [40 ft-lb]



⚠ CAUTION ⚠

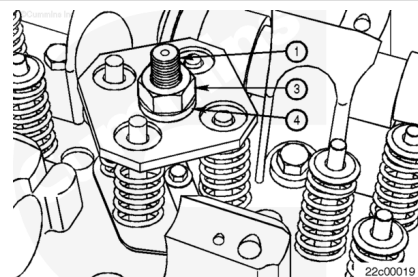
The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in damage to the engine and/or tool.



Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.

Install the washer (4) and forcing nut (3) on the replacer screw (1).

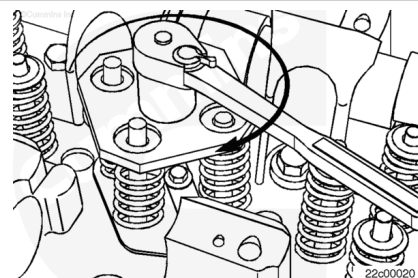
Tighten finger-tight.



⚠ WARNING ⚠

Eye protection should be worn during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.

Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

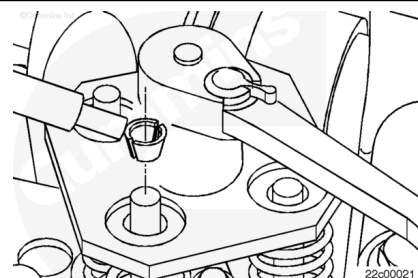


⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.

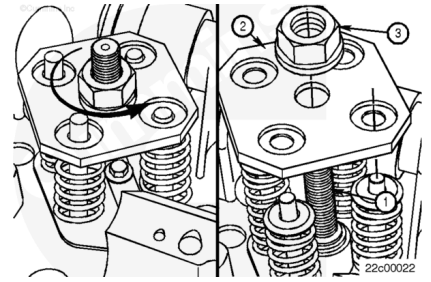
Continue turning the forcing nut (3) **clockwise** until the valve collets can be removed with a magnetic tool.

Discard the valve collets.



⚠ WARNING ⚠

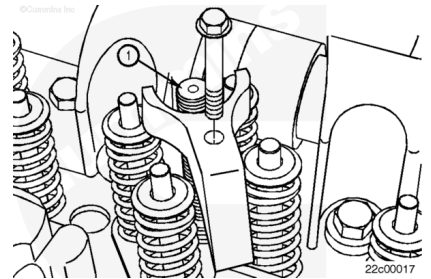
Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.



Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).

Remove the replacer screw (1) by removing the injector hold down bolt and injector hold down clamp.



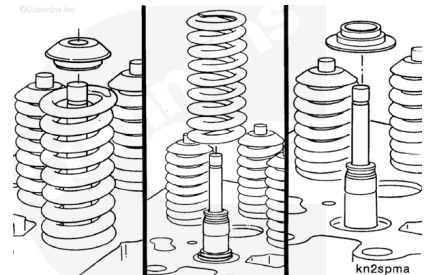
Remove the valve spring retainer.

Remove the valve spring.

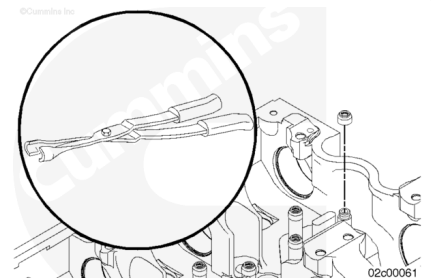
If fitted, remove the valve spring guide.

Repeat the above steps for each cylinder as needed.

Note : Valve guides that do not have the groove must have the valve spring guides removed for use of the installation tool.



Where guide mounted valve seals are fitted, use the boot pliers, Part Number 3163293, to remove the valve guide seals.



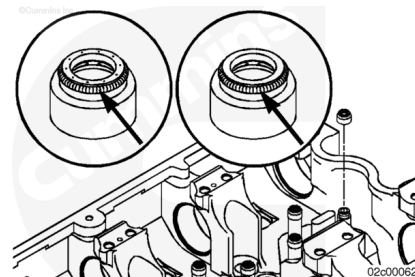
Install

Grooved Valve Guide Seals:

Note : The seals that mate with the grooved valve guides have two iterations.

1. Engines built from 15 May 2000 to the end of Feb 2002 had separate intake and exhaust valve guide seals which were color coded. Intake valve guide seals had a brass-colored garter spring and three raised dots molded into the rubber; the exhaust valve guide seals had a silver-colored garter spring and **no** raised dots.
2. Engines built after March 2002 had **only** one valve guide seal for both intake and exhaust valve guides.

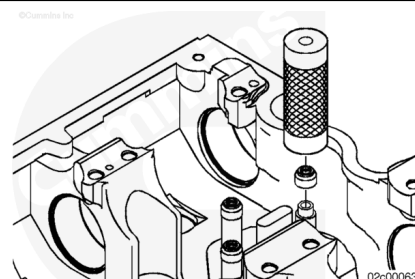
Install the intake and exhaust valve guide seals by snapping the seal by hand into the groove on the valve guide.



Non-Grooved Valve Guide Seals:

Note : Make sure the valve spring guides are removed before installing the non-grooved valve guide seals.

Use an unmodified valve seal installer, Part Number 3163294, to install the valve guide seals. The seal installer **must** make contact with the cylinder head when installed. This correctly locates the valve guide seal.

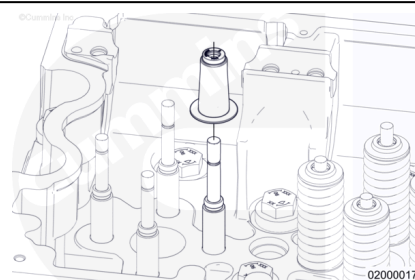


⚠ CAUTION ⚠

To avoid engine and/or tool damage, do not use a service tool to install valve stem seals with an integrated valve spring guide.

⚠ CAUTION ⚠

To avoid engine damage, ensure valve spring guides are removed before attempting to install a valve stem seal.



Valve Stem Seal with Integrated Valve Spring Guide:

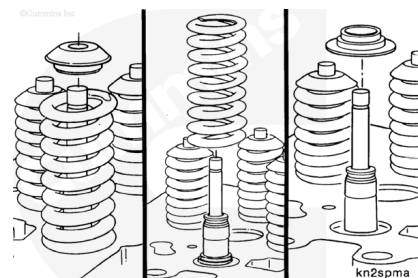
- Valve stem seals with the integrated valve spring guide are installed by hand without the use of an installation tool.
- Locate the valve stem seal over the valve stem and guide and firmly push into place by hand until the valve spring guide is seated firmly against the cylinder head.

Install the valve spring retainer.

Install the valve spring.

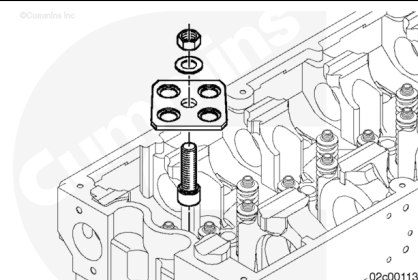
If required, install the valve spring guide.

Repeat the above steps for each cylinder as needed.



⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if improperly released. To reduce the possibility of eye injury, wear safety glasses with side shields. Before releasing the valve spring tension, check that the collets are properly seated.

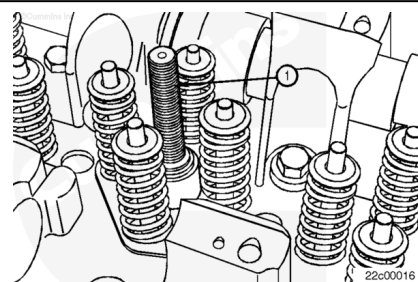


⚠ CAUTION ⚠

**Do not use an impact tool for any part of this procedure.
Doing so can damage the tool and/or the engine.**

Use the four valve spring compressor, Part Number 4919358, to compress the valve springs.

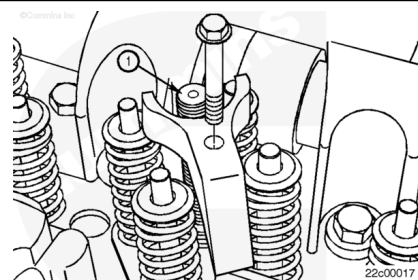
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

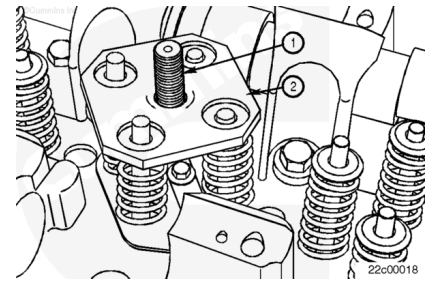
Tighten the capscrew.

Torque Value: 54 n·m [40 ft-lb]



⚠ CAUTION ⚠

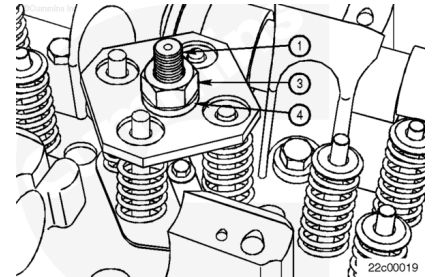
The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in damage to the engine and/or tool.



Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.

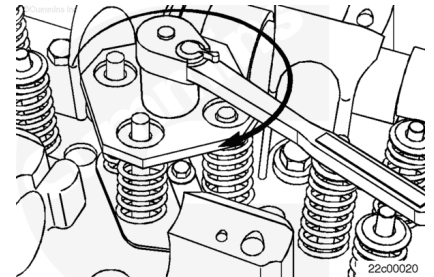
Install the washer (4) and forcing nut (3) on the replacer screw (1).

Tighten the nut finger-tight.



⚠ WARNING ⚠

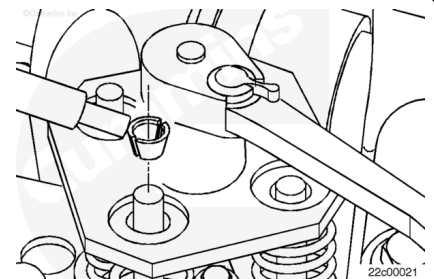
Eye protection should be worn during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.



Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.



Continue turning the forcing nut (3) **clockwise** until the valve collets can be installed.

Always use new collets when rebuilding the cylinder head.

Install the new valve spring retainer collets.

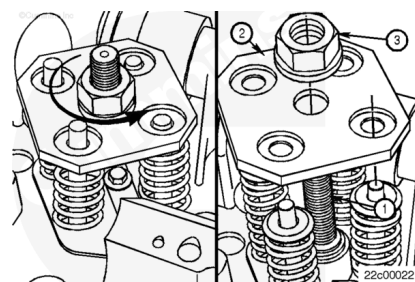
Inspect the collets for proper seating and installation.

⚠ WARNING ⚠

Eye protection should be worn during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.

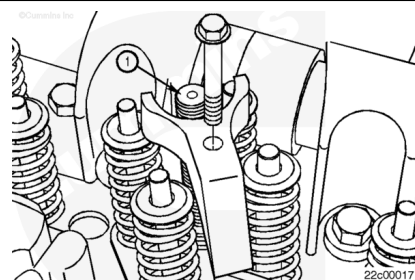
Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).



Remove the replacer screw (1) by removing the injector hold-down bolt and injector hold-down clamp.

Repeat the above steps for each cylinder as needed.



Finishing Steps

On-Engine Repair

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html>)
- Install the rocker lever shaft assemblies. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>).
- Connect the engine brake solenoid wiring harness. Refer to Procedure 020-015 in Section 20. (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>)
- Adjust the overhead set as needed. Refer to Procedure 003-004 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html>).
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- If failure resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)
- Operate the engine and check for leaks and proper operation.

Off-Engine Repair

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html>)
- If failure resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)
- Operate the engine and check for leaks and proper operation.